

**NYU****TANDON SCHOOL
OF ENGINEERING****Term: Fall 2026****Department of Mechanical and Aerospace Engineering**

ME-GY 4214 Finite Element Design, Modeling and Analysis

Thursday, 6-7:50 pm, Location: 6 MTC 473**Instructor: Dr. Nikhil Gupta****Email: ngupta@nyu.edu**

The analysis of complex static and dynamic problems involves three steps: selection of a mathematical model; analysis of the model; interpretation of the predicted response. The course deals with deriving analytical solutions and comparing them with Finite Element Analysis results. Students are required to use state-of-the-art commercial software.

Week	Date		Topic
Week 1	9/3/2026		Introduction to Finite Element Analysis (FEA)
Week 2	9/10/2026		FEM steps and spring element
Week 3	9/17/2026		Matrix algebra and spring element examples
Week 4	9/24/2026		2D problems and bar elements
Week 5	10/1/2026	Midterm 1	2D problems and bar elements
Week 6	10/8/2026		Bar element examples
Week 7	10/15/2026		Beam element
Week 8	10/22/2026	Midterm 2	Beam element
Week 9	10/29/2026		Energy method
Week 10	11/5/2026		2D triangle element
Week 11	11/12/2026	Midterm 3	2D triangle element
Week 12	11/19/2026		Practical modeling techniques
Week 13	12/3/2026		Machine Learning in FEM
Week 14	12/10/2026	Midterm 4	Review
Final Exam	12/17/2026		Final Exam

Learning Objectives	Learn the modelling procedures implemented in ANSYS to solve problems from various disciplines of mechanical engineering.																							
Learning Outcomes	Students will able to: • Implement the principal elements using the ANSYS Workbench • Construct the finite element model and analyze the complex problem in mechanical structure and heat transfer using ANSYS • Learn the impact of FEM on engineering domain to solve real-world issues • Be able to use engineering judgment to draw conclusions.																							
Grading scheme	<table><tr><td colspan="3">Lecture Class</td></tr><tr><td></td><td>Mid-term exams* (4 given, best 3 counted)</td><td>30%</td></tr><tr><td></td><td>Final exam</td><td>20%</td></tr><tr><td>Lab</td><td>4 lab assignments</td><td>15%</td></tr><tr><td></td><td>Lab project</td><td>30%</td></tr><tr><td></td><td>Attendance</td><td>5%</td></tr><tr><td></td><td>Total</td><td>100%</td></tr></table>			Lecture Class				Mid-term exams* (4 given, best 3 counted)	30%		Final exam	20%	Lab	4 lab assignments	15%		Lab project	30%		Attendance	5%		Total	100%
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***Midterm exams will be 45 min – 1 hr long.**